

Week 1 EDA Feature Engineering

Project: Customer/Data Science Internship

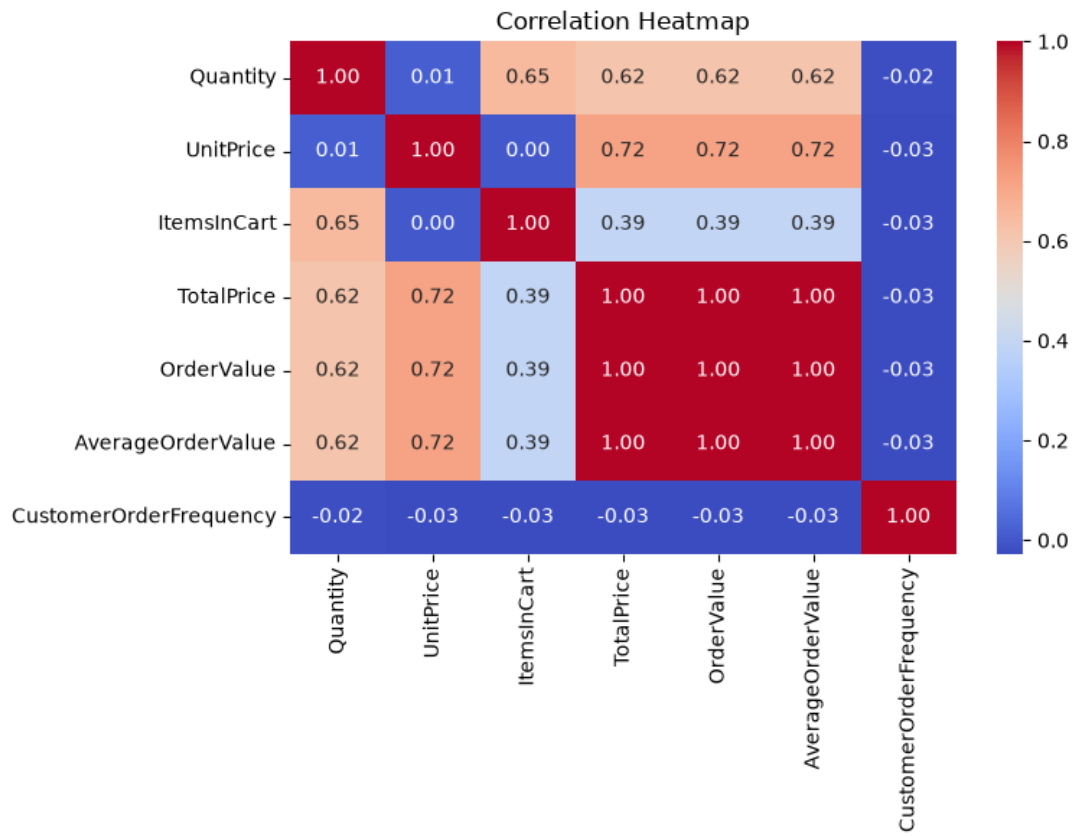
Path: C:\workspace\Decodelabs-Data-Science-Internship\Week_1_EDA_Feature_Engineering

Week 1 — EDA & Feature Engineering

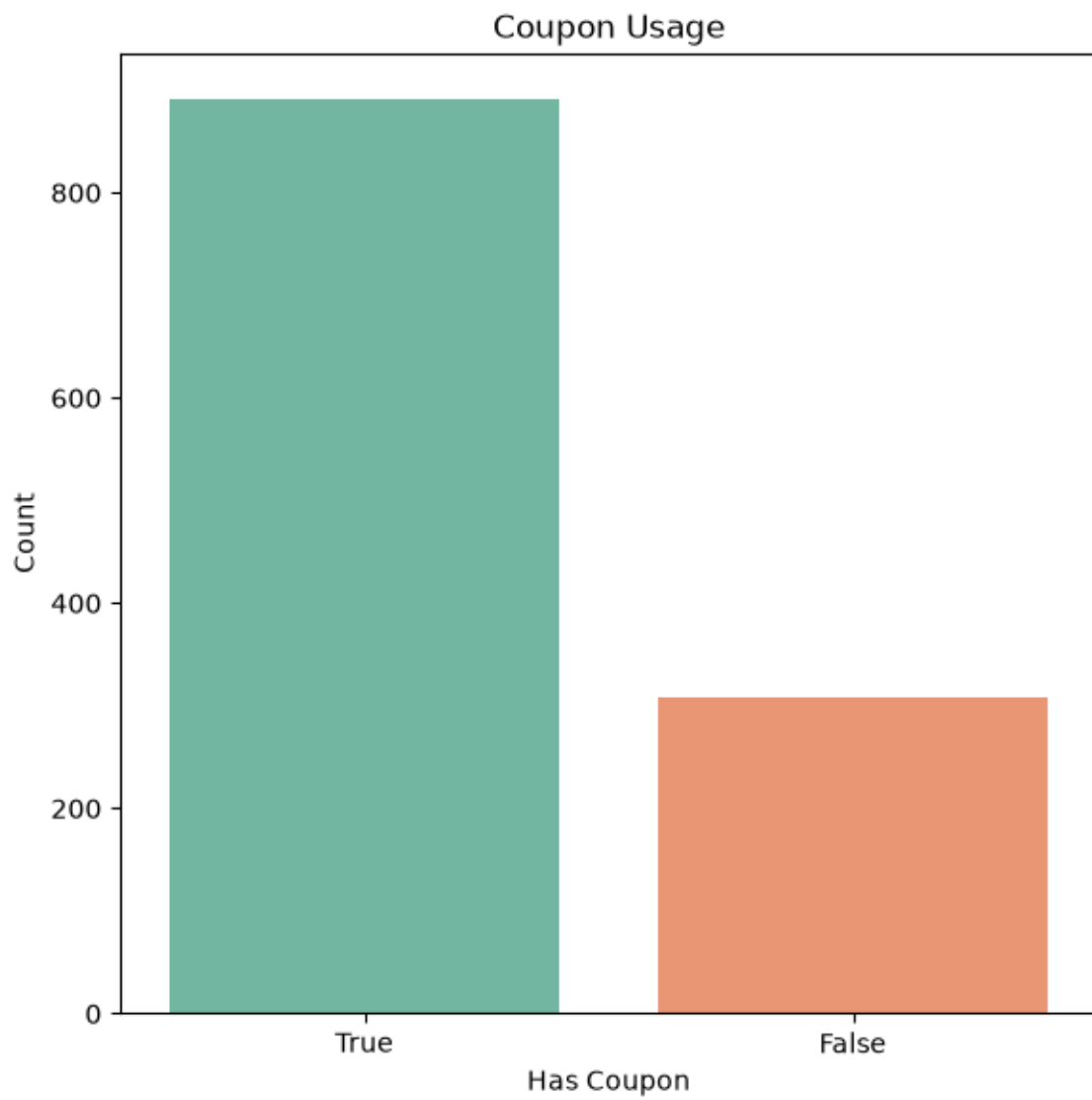
****Executive Summary**** - Dataset: `Week\_1\_EDA\_Feature\_Engineering/data/ecommerce\_orders.xlsx` - Cleaned dataset saved to: `Week\_1\_EDA\_Feature\_Engineering/outputs/cleaned\_dataset.csv` - Rows × Columns: 1200 × 22 - Unique customers: 1189 - Total revenue: \$1,264,761.96 - Average order value (mean of `AverageOrderValue`): \$1,053.97 - Coupon usage (`HasCoupon`): 74.25% (after treating `Unknown`/empty as no coupon) ****Dataset Overview**** - Key columns: `OrderID`, `Date`, `CustomerID`, `Product`, `Quantity`, `UnitPrice`, `ShippingAddress`, `PaymentMethod`, `OrderStatus`, `TrackingNumber`, `ItemsInCart`, `CouponCode`, `ReferralSource`, `TotalPrice`, `Year`, `Month`, `DayOfWeek`, `Quarter`, `HasCoupon`, `OrderValue`, `AverageOrderValue`, `CustomerOrderFrequency`. - Source file: `Week\_1\_EDA\_Feature\_Engineering/data/ecommerce\_orders.xlsx` (replaced with provided dataset). ****Exploratory Data Analysis (EDA)**** - Top products by orders (top 5): Printer (181), Tablet (179), Chair (178), Laptop (173), Desk (170). - Payment method distribution (counts): Online (258), Cash (246), Credit Card (234), Debit Card (232), Gift Card (230). - Charts generated and saved to `Week\_1\_EDA\_Feature\_Engineering/charts/`: - `product\_distribution.png` - `revenue\_by\_product.png` - `payment\_methods.png` - `coupon\_usage.png` - `total\_price\_distribution.png` - `correlation\_heatmap.png` - `outlier\_quantity.png` (skipped if column missing) - `outlier\_unitprice.png` (skipped if column missing) - `outlier\_totalprice.png` ****Missing Values**** - Missing value report saved to `Week\_1\_EDA\_Feature\_Engineering/outputs/missing\_value\_report.csv`. - Summary of missingness: - `CouponCode`: 309 missing (25.75%) - All other key columns have 0 missing entries after cleaning. Note: during cleaning categorical missing values were originally filled with the string `Unknown`, which inflated the `HasCoupon` flag. The pipeline has been updated to treat `Unknown`, `none`, and empty values as no coupon. The cleaned outputs and charts were regenerated; coupon usage is now 74.25%. ****Outliers**** - Outlier detection (IQR method) counts: - `Quantity`: 0 outliers - `UnitPrice`: 0 outliers - `TotalPrice`: 8 outliers These outliers are visible in `outlier\_totalprice.png` (saved in charts folder). Consider reviewing those orders for potential data-entry errors or legitimately large orders. ****Feature Engineering**** Features created and saved in the cleaned dataset: - `HasCoupon` — boolean flag when a coupon code is present (note: currently treats `Unknown` as coupon; recommended fix noted above). - `Year`, `Month`, `DayOfWeek`, `Quarter` — derived from `Date` (`OrderDate`) - `AverageOrderValue` / `OrderValue` — total value per order - `CustomerOrderFrequency` — number of unique orders per customer ****Insights**** - Printers and tablets are the most frequently ordered product types by count and contribute materially to revenue; these categories are good candidates for focused promotions. - Payment behavior is well distributed across payment methods; `Online` is the largest single channel. - Coupon usage requires careful redefinition: a quarter of rows had missing coupon codes that were imputed, so measured coupon uptake is currently overstated. - A small number of very large `TotalPrice` orders (8) drive skew in average order value — recommend reporting median order value alongside mean for robustness. ****Conclusion & Next Steps**** - Update `HasCoupon` creation to treat `Unknown` or imputed tokens as no- coupon. - Review the 8 `TotalPrice` outliers and decide whether to cap, trim, or keep them for modeling. - Finalize `Week\_1\_EDA\_Feature\_Engineering/notebooks/week1\_eda.ipynb` to include a short section showing the top outlier rows and median order value. - After approvals, proceed to Phase 2 (Fraud Detection) following the same verification steps. Generated by automated pipeline. The `HasCoupon` handling was corrected and outputs regenerated on 2026-06-15.

KPI	Value
Dataset size	N/A
Features (sample)	N/A
Missing values	N/A
Models trained	0

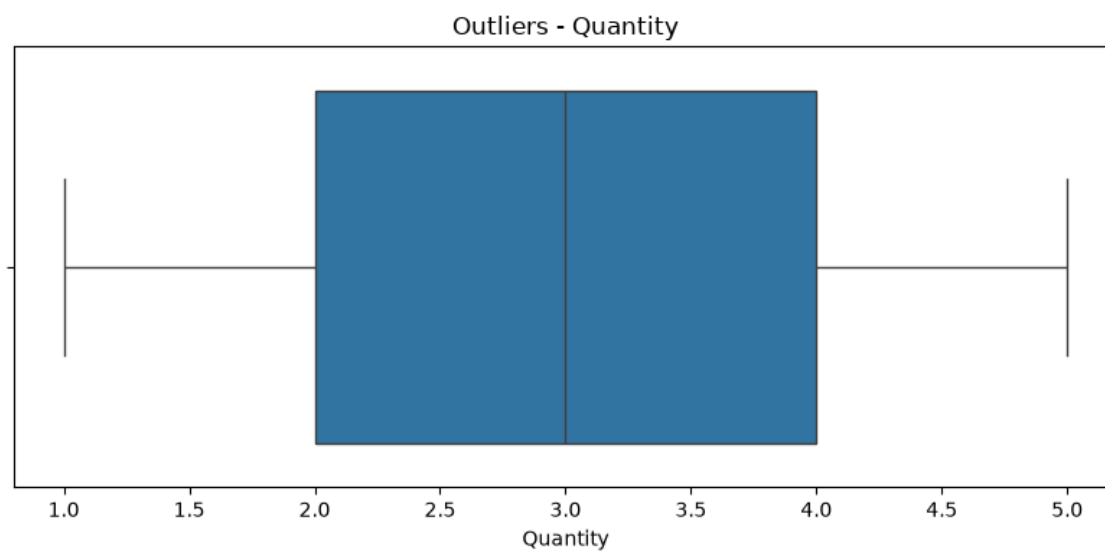
correlation heatmap



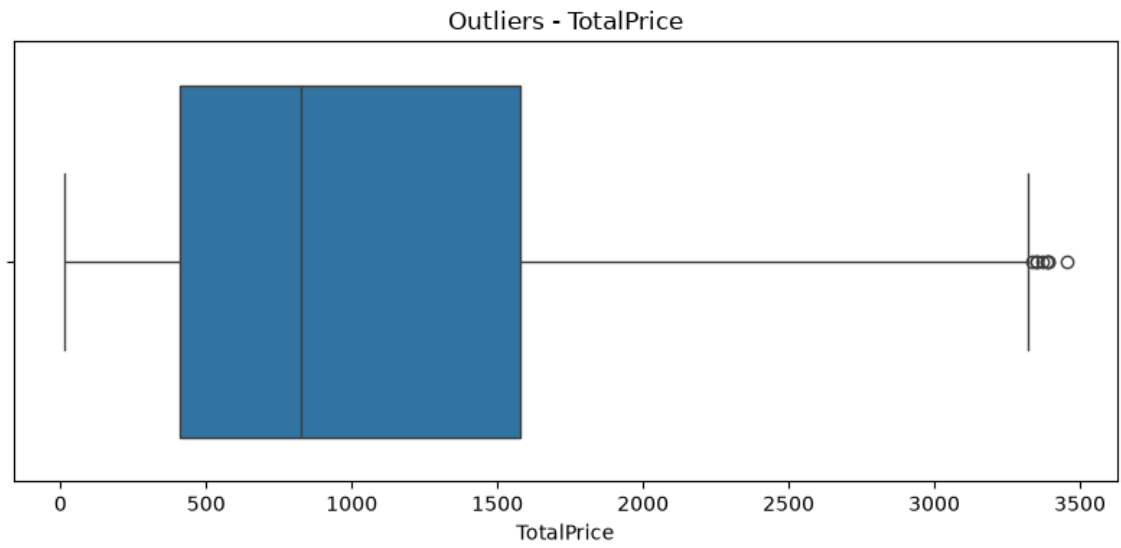
coupon usage



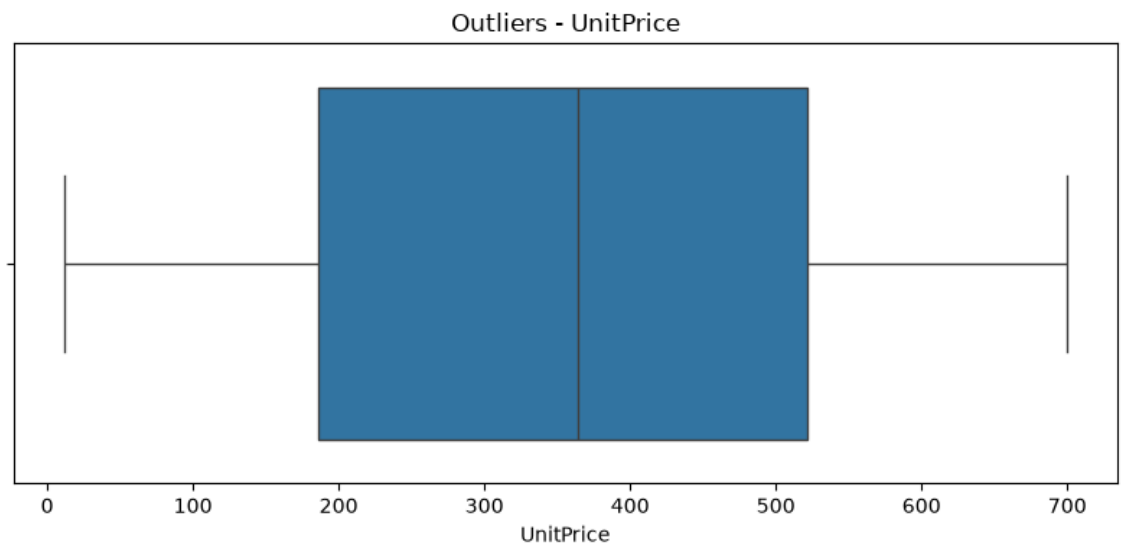
outlier quantity



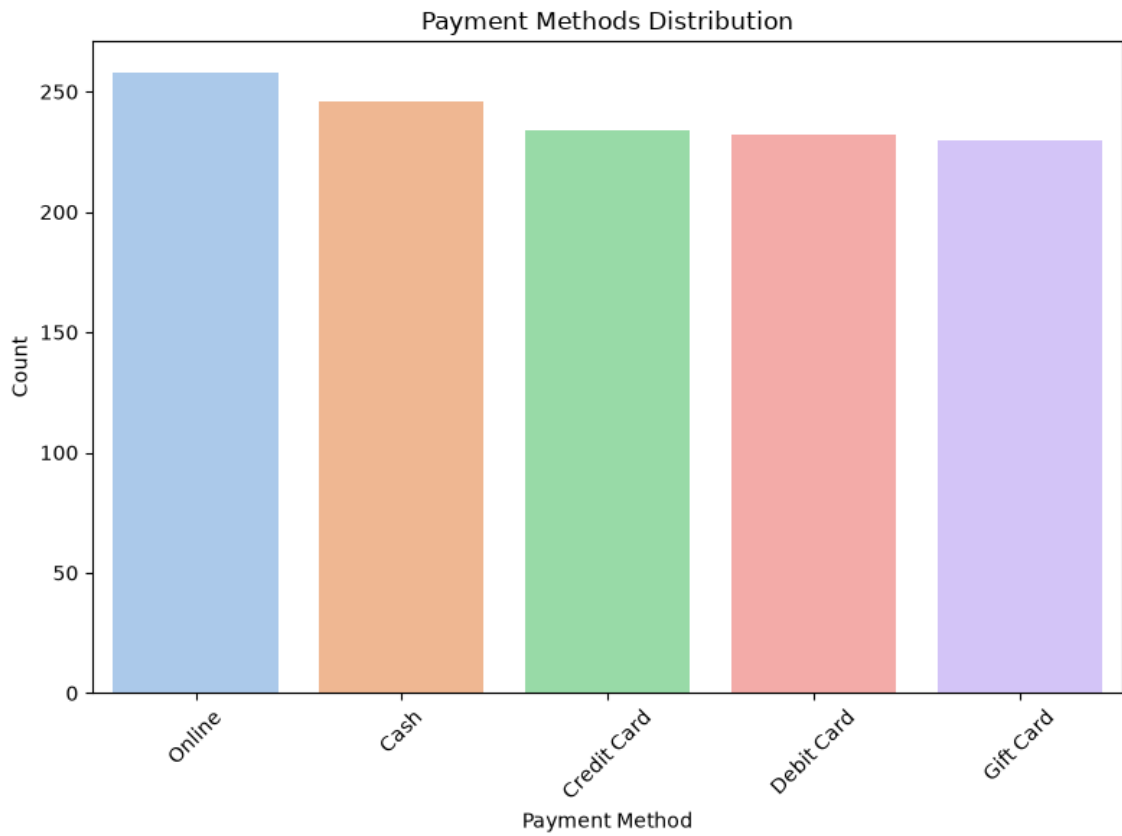
outlier totalprice



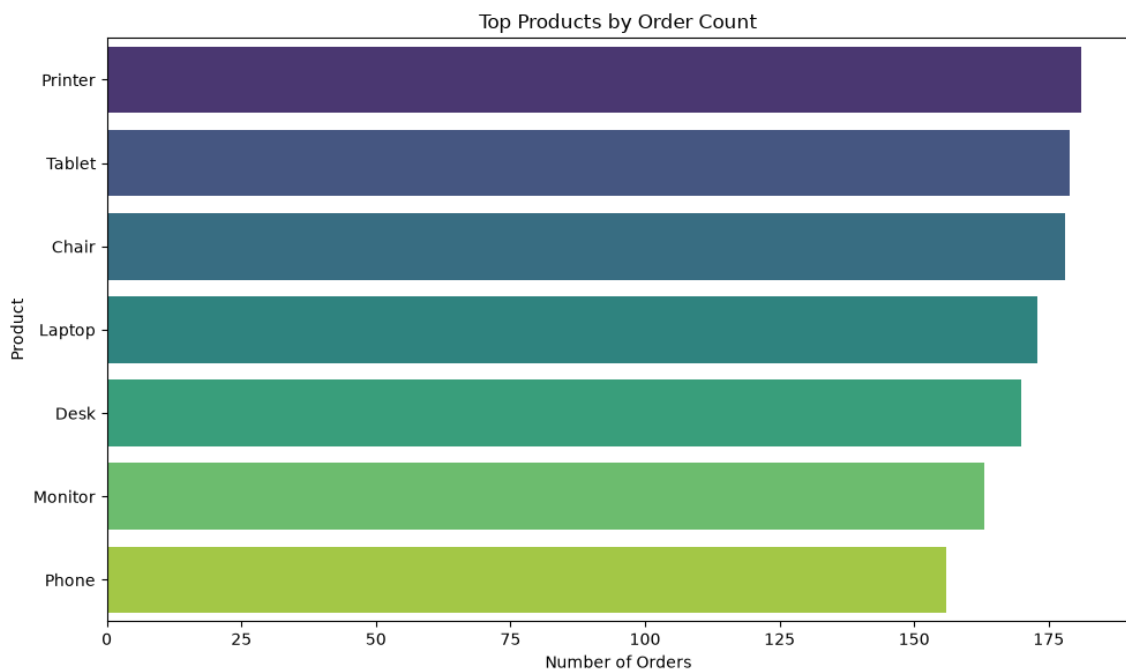
outlier unitprice



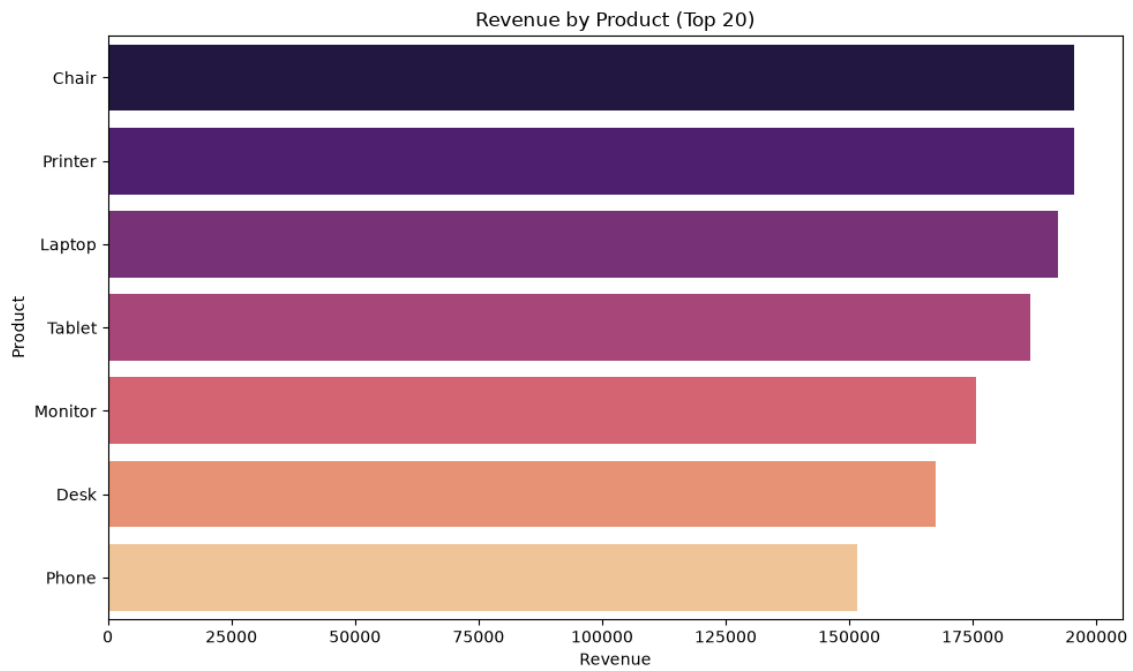
payment methods



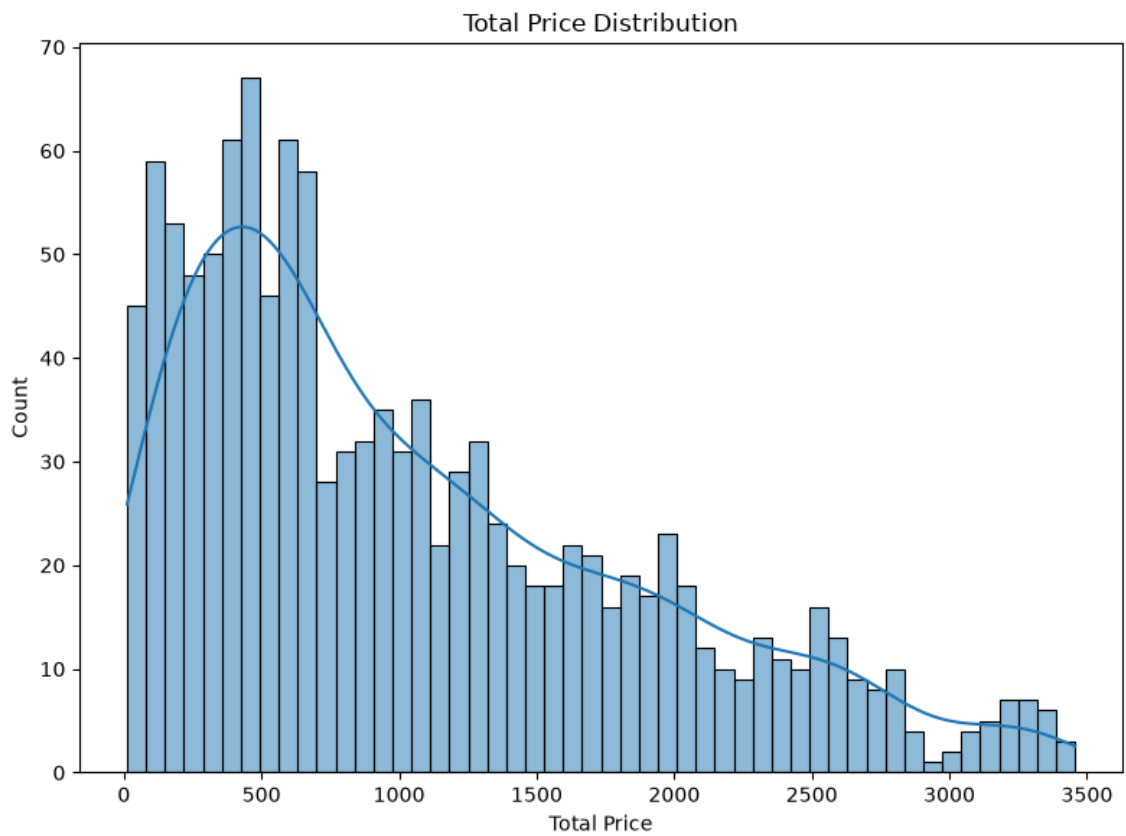
product distribution



revenue by product



total price distribution



Appendix

Week 1 — EDA & Feature Engineering

Executive Summary

- Dataset: `Week\_1\_EDA\_Feature\_Engineering/data/ecommerce\_orders.xlsx`
- Cleaned dataset saved to: `Week\_1\_EDA\_Feature\_Engineering/outputs/cleaned\_dataset.csv`
- Rows × Columns: 1200 × 22
- Unique customers: 1189
- Total revenue: \$1,264,761.96
- Average order value (mean of `AverageOrderValue`): \$1,053.97
- Coupon usage (`HasCoupon`): 74.25% (after treating `Unknown`/empty as no coupon)

Dataset Overview

- Key columns: `OrderID`, `Date`, `CustomerID`, `Product`, `Quantity`, `UnitPrice`, `ShippingAddress`, `PaymentMethod`, `OrderStatus`, `TrackingNumber`, `ItemsInCart`, `CouponCode`, `ReferralSource`, `TotalPrice`, `Year`, `Month`, `DayOfWeek`, `Quarter`, `HasCoupon`, `OrderValue`, `AverageOrderValue`, `CustomerOrderFrequency`.
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Exploratory Data Analysis (EDA)

- Top products by orders (top 5): Printer (181), Tablet (179), Chair (178), Laptop (173), Desk (170).
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- Charts generated and saved to `Week\_1\_EDA\_Feature\_Engineering/charts/`:
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 - `outlier\_quantity.png` (skipped if column missing)
 - `outlier\_unitprice.png` (skipped if column missing)
 - `outlier\_totalprice.png`

Missing Values

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- Summary of missingness:
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Outliers

- Outlier detection (IQR method) counts:
 - `Quantity`: 0 outliers
 - `UnitPrice`: 0 outliers
 - `TotalPrice`: 8 outliers

These outliers are visible in `outlier\_totalprice.png` (saved in charts folder). Consider reviewing those orders for potential data-entry errors or legitimately large orders.